House Price Analysis and Prediction

Project by: Muhammad Farooq

Program: Digital Empowerment Network - Data Science

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Mentor: Ali Mohiuddin Khan

- **Project Overview**

This project focuses on analyzing real estate property listings to understand key factors affecting house prices and building machine learning models for accurate price prediction. The dataset includes features such as property type, location, price, size, and the number of rooms. The goal is to process the data, extract meaningful insights, handle outliers, and build models to predict future house prices.

- **Objectives**

- Clean and explore the dataset to understand house price trends.

- Engineer new features such as house age and price per square foot.

- Identify and handle outliers.

- Train and evaluate different machine learning models.

- Predict prices for hypothetical scenarios.

- **Methodology**

1. Data Cleaning:

  - Removed invalid or zero prices.

  - Replaced missing values in 'agency' and 'agent'.

  - Ensured valid ranges for latitude, longitude, and bedroom/bath counts.

  - Removed duplicates and standardized categorical fields.

2. Feature Engineering:

- Created new features: house_age, price_per_sqft, baths_per_bedroom, and bedrooms_per_floor.

- Converted date and area values for calculations.

- Applied one-hot encoding to categorical variables.

3. Outlier Handling:

- Used Winsorization and IQR method to handle extreme values in numerical columns.

- Identified outliers using Z-score method.

4. Visualization:

- Used boxplots, scatter plots, and heatmaps to analyze feature relationships and distributions.

5. Predictive Modeling:

- Used three models: Linear Regression, Random Forest, and Gradient Boosting.

- Split data into training and testing sets (80/20).

- Evaluated models using MAE, MSE, and R^2 score.

6. Future Price Prediction:

- Predicted prices using trained models for new hypothetical properties.

- **Challenges**

- Handling inconsistent or missing values in a large dataset (168,000+ rows).

- Feature conversion (especially area formats and date formats).

- Ensuring meaningful and interpretable feature engineering.

- Balancing model accuracy and interpretability.

- **Conclusion**

- Random Forest performed best with an R^2 score of 0.59, followed by Gradient Boosting and Linear Regression.

- Important features impacting house price include area size, number of bedrooms/baths, location coordinates, and property age.

- The model can assist real estate agencies and buyers in making data-driven pricing decisions.

- **Recommendations:**

- Include additional features like amenities, nearby facilities, or construction year.

- Use geospatial clustering for more location-specific insights.

- Try advanced models like XGBoost or Deep Learning for further improvement.